Study Guide for Population Growth and Demographics

1. A, D, and E are density dependent
   B, C, and F are density independent
2. Answer is A
3. Answer is F
4.
   a. $R_0 = 2.41$ (taken from the table, no calculation necessary)
   b. G = 20.37/2.41 = 8.45
   c. r = (ln 2.41)/8.45 = 0.104
5. Answer is E
6.
   a. Left
   b. Dashed line
   c. Dashed line
   d. Solid Line

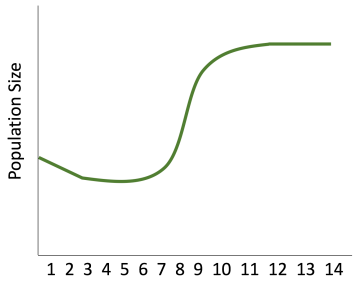
7. Answer is D
8.

Age class	Average Age in Years	Proportion Surviving (l_x)	Age-specific Fecundity (m_x)	Lxmx	Xlxmx
hatchlings	0	1.0	0	0	0
juvenile	1	0.75	0	0	0
Young adult	5	0.50	10	5	25
Adult	7	0.25	12	3	21
Old Adult	10+	0.01	0	0	0
TOTALS				8	46

   a.
      i. 8
      ii. 5.75
      iii. 0.36
   b. Increasing; the R_0 value is greater than 1 and the *r* value is greater than 0, both of which indicate a population that is increasing.

9.
   a. 676.39
   b. 563.75
   c. 2.31
   d. 17.33

10.



11.

a. 823
b. 225
c. Both populations have a very similar increase in terms of numbers of individuals (A increases by 23 individuals and B increases by 25); however because Population B is so much smaller, the proportional increase is larger (ie, 25/200 is larger than 23/800).
d. Yes, though we would need more complex equations to represent that situation very well. If a population overshoots its carrying capacity, it will most likely experience a negative growth rate until it goes back below the carrying capacity. Our equation doesn't 'allow' a population below K to overshoot, though it does predict a decline in a population size for a population that is larger than its carrying capacity. *Note: this is a good example of where mathematical predictions can diverge from biological reality. While overshooting carrying capacity is biologically possible, the simplicity of the equation we use in this class does not effectively model how that occurs. To do so, we would need a more complex mathematical model.*

12. R_o is the average number of offspring an individual in the population is expected to have over the course of their entire life, while *r* is the growth rate of the population as a whole. When R_o is greater than 1, each individual is, on average, more than replacing itself in the next generation, and the population will be increasing, with a positive *r*. When R_o is less than 1, not every individual is, on average, replacing itself in the next generation, and the population will be decreasing, with a negative r. When R_o is equal to 1, each individual is on average just replacing itself in the next generation, and the population will be stable, with r =0. I keep saying 'on average' in these sentences to point out that the variation in reproductive output between individuals can be very high. For example, if offspring number ranges from 0 to over 100, an R_0 value of greater than 1 can be achieved even if many individuals are not reproducing at all.

13. Scientists can estimate N_0 through direct counting or mark-recapture studies (including photographic mark recapture). The growth rate (*r*) can be estimated by following a population through time and counting individuals, or by using life table analysis to estimate r from R_0. There are other mechanisms for these estimates, but these are the ones we discussed in lecture.

14.

a. Use the logistic growth equation, fill in the values you know (N_t, N_0, K and t) and solve for r. Answer is -0.234

$$N_t = \frac{K}{1+\left(\frac{K-N_0}{N_0}\right)e^{-rt}}$$

$$200 = \frac{500}{1+\left(\frac{500-437}{437}\right)e^{-r10}}$$

$$200(1+0.144e^{-r10}) = 500$$

$$1+0.144e^{-r10} = 500/200 = 2.5$$

$$0.144e^{-r10} = 2.5-1$$

$$e^{-r10} = 1.5/0.144 = 10.42$$

$$-r10(\ln e) = \ln(10.42) = 2.34$$

$$-r = 2.34/10 = 0.234$$

$$r = -0.234$$

b. We can estimate r with $\ln(R_0)/G$, so fill in the numbers you know and solve for R_0
-0.234 = (ln (R_0))/1 = ln(R_0)
e^(-0.234) = e^(ln(R_0)) ← *remember that e^ln(x) = x*
e^(-0.234) = R_0 = 0.7914

15. There are lots of possible explanations. Geckos molt (shed their outer layer of skin) which means any geckos that molted between your tagging and second observation would no longer have their paint tags. Even in animals that don't molt, the tags can fall off. Perhaps the paint tags made geckos more visible to predators and all the geckos you tagged were eaten. Perhaps the population of geckos in this habitat is so large that only 15 tagged individuals is too small a sample to observe in the second visit. Geckos aren't migratory, but in species that are, migration out of the sampling area could also explain this result.